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September 6, 2024

U.S. Nuclear Regulatory Commission
ATTN: Document Control Desk
Washington, DC 20555

Calvert Cliffs Nuclear Power Plant, Unit Nos 1 and. 2
Renewed Facility Operating License Nos. DPR-53 and DPR-69
NRC Docket Nos. 50-317 and 50-318

Subject: Licensee Event Report 2024-003, Revision 00
1A Diesel Generator Inoperable Due to Potential Transformer Failure

The attached report is being sent to you as required by 10 CFR 50.73.

There are no regulatory commitments contained in this correspondence.

Should you have questions regarding this report, please contact Mr. Larry D. Smith at (667) 313-6503.

Respectfully,

Christopher J. Smith
Plant Manager

CJS/TEL/ak

Attachment: LER 317-2024-003, Rev 00

cc: NRC Project Manager, Calvert Cliffs
NRC Regional Administrator, Region I
NRC Resident Inspector, Calvert Cliffs
S. Seaman, DNR

NRC FORM 366 (04-02-2024)	U.S. NUCLEAR REGULATORY COMMISSION	APPROVED BY OMB: NO. 3150-0104	EXPIRES: 04/30/2027

LICENSEE EVENT REPORT (LER)

(See Page 2 for required number of digits/characters for each block)

(See NUREG-1022, R.3 for instruction and guidance for completing this form http://www.nrc.gov/reading-rm/doc-collections/nuregs/staff/sr1022/r3/)

1. Facility Name		2. Docket Number	3. Page
Calvert Cliffs Nuclear Power Plant, Unit 1	☒ 050 ☐ 052	05000317	1 OF 4

4. Title

1A Diesel Generator Inoperable Due to Potential Transformer Failure

5. Event Date			6. LER Number			7. Report Date		
Month	Day	Year	Year	Sequential Number	Revision No.	Month	Day	Year
07	08	2024	2024	- 003 -	0	09	06	2024

8. Other Facilities Involved		
Facility Name		Docket Number
Calvert Cliffs Nuclear Power Plant, Unit 2	☒ 050	05000318
N/A	☐ 052	

9. Operating Mode	10. Power Level
1 – Power Operation	100

11. This Report is Submitted Pursuant to the Requirements of 10 CFR §: *(Check all that apply)*

10 CFR Part 20	☐ 20.2203(a)(2)(vi)	**10 CFR Part 50**	☐ 50.73(a)(2)(ii)(A)	☐ 50.73(a)(2)(viii)(A)	☐ 73.1200(a)
☐ 20.2201(b)	☐ 20.2203(a)(3)(i)	☐ 50.36(c)(1)(i)(A)	☐ 50.73(a)(2)(ii)(B)	☐ 50.73(a)(2)(viii)(B)	☐ 73.1200(b)
☐ 20.2201(d)	☐ 20.2203(a)(3)(ii)	☐ 50.36(c)(1)(ii)(A)	☐ 50.73(a)(2)(iii)	☐ 50.73(a)(2)(ix)(A)	☐ 73.1200(c)
☐ 20.2203(a)(1)	☐ 20.2203(a)(4)	☐ 50.36(c)(2)	☐ 50.73(a)(2)(iv)(A)	☐ 50.73(a)(2)(x)	☐ 73.1200(d)
☐ 20.2203(a)(2)(i)	**10 CFR Part 21**	☐ 50.46(a)(3)(ii)	☐ 50.73(a)(2)(v)(A)	**10 CFR Part 73**	☐ 73.1200(e)
☐ 20.2203(a)(2)(ii)	☐ 21.2(c)	☐ 50.69(g)	☐ 50.73(a)(2)(v)(B)	☐ 73.77(a)(1)	☐ 73.1200(f)
☐ 20.2203(a)(2)(iii)		☐ 50.73(a)(2)(i)(A)	☐ 50.73(a)(2)(v)(C)	☐ 73.77(a)(2)(i)	☐ 73.1200(g)
☐ 20.2203(a)(2)(iv)		☒ 50.73(a)(2)(i)(B)	☐ 50.73(a)(2)(v)(D)	☐ 73.77(a)(2)(ii)	☐ 73.1200(h)
☐ 20.2203(a)(2)(v)		☐ 50.73(a)(2)(i)(C)	☐ 50.73(a)(2)(vii)		

☐ **OTHER** (Specify here, in abstract, or NRC 366A).

12. Licensee Contact for this LER

Licensee Contact	Phone Number (Include area code)
Travis E. Lefton, Principal Regulatory Engineer	667-313-6402

13. Complete One Line for each Component Failure Described in this Report

Cause	System	Component	Manufacturer	Reportable to IRIS		Cause	System	Component	Manufacturer	Reportable to IRIS
B	EK	XPT	G090	Y						

14. Supplemental Report Expected		15. Expected Submission Date	Month	Day	Year
☒ No	☐ Yes (If yes, complete 15. Expected Submission Date)				

16. Abstract (Limit to 1326 spaces, i.e., approximately 13 single-spaced typewritten lines)

On July 2, 2024, the Main Control Room received alarm “4KV BUS U/V, 480V BUS U/V, 125V DC,” Window AA06. Initial investigation using the Alarm Response Manual (ARM) did not identify any abnormal degraded conditions and the 1A Emergency Diesel Generator (EDG) was immediately determined to be operable. Follow-up investigation and troubleshooting identified a failed potential transformer associated with the 17 4KV bus protective relays, which would result in the 1A EDG output breaker tripping open on overcurrent during load sequencing in an emergency scenario. The 1A EDG was promptly declared inoperable on July 8, 2024, and the station entered Technical Specification Condition 3.8.1.B for Unit 1 and 3.8.1.E for Unit 2. After entering the Technical Specification Conditions the site took all requisite Required Actions. The potential transformer was replaced on July 10, 2024, and following a successful surveillance run, the 1A EDG was declared operable on July 11, 2024.

Prior to entering the Technical Specification Conditions and taking the associated requisite Required Actions, in the period between July 2, 2024, and July 8, 2024, the station exceeded the completion time for Technical Specification LCO 3.8.1 for Unit 1 and 3.8.1 for Unit 2; therefore, this event is reportable under 10CFR50.73(a)(2)(i)(B) as an operation or condition which was prohibited by the plant's Technical Specifications.

1. FACILITY NAME		2. DOCKET NUMBER	3. LER NUMBER YEAR	SEQUENTIAL NUMBER	REV NO.
Calvert Cliffs Nuclear Power Plant, Unit 1	■ 050 / ☐ 052	05000317	2024	- 003	- 00

NARRATIVE

PLANT AND SYSTEM IDENTIFICATION

Calvert Cliffs Nuclear Power Plant, Units 1 and Unit 2 are Combustion Engineering Pressurized Water Reactors with a licensed maximum power level of 2737 megawatts thermal, respectively. The Energy Industry Identification System code used in the text is identified as EK.

A. CONDITION PRIOR TO EVENT

Unit: 1
Date: July 2, 2024
Power level: 100
Mode: Unit 1 was in Mode 1 when the condition was discovered

Unit: 2
Date: July 2, 2024
Power level: 100
Mode: Unit 2 was in Mode 1 when the condition was discovered

B. DESCRIPTION OF EVENT

On 7/2/24 an alarm, “4kV Bus U/V, 480 V Bus U/V, 125 VDC,” was received in the Control Room. This is a general alarm associated with the 17 4kV bus, the 07 480V bus and the 125VDC systems which support operation of the 1A Emergency Diesel Generator (EDG). Investigation using the Alarm Response Manual (ARM) in the field and in the Control Room did not identify an apparent cause of the alarm. Information provided to the Unit Supervisor was that the failure involved the alarm circuit and did not impact operability of the 1A EDG. Initial assessment performed by Shift SROs was that the 1A EDG remained operable.

Investigations into the 1A EDG alarm were performed by Fix-it-Now (FIN) personnel on 7/2 and 7/3/24. The potential transformer (PT) supporting the protection circuit and synchronizing circuit was identified as the failed component and the cause for the alarm. On 7/8/24 Electrical Maintenance and Engineering identified the voltage input to the voltage-restraint overcurrent device as being impacted by the failed PT which would cause an overcurrent trip of the 1A EDG output breaker during an accident scenario. Shift Senior Reactor Operators (SROs) reviewed the new information and promptly declared the 1A EDG inoperable and entered Technical Specification (TS) Condition 3.8.1.B for Unit 1 and LCO 3.8.1.E for Unit 2.

C. DATES AND APPROXIMATE TIMES OF MAJOR OCCURRENCES

<u>Event Timeline:</u>

7/02/2024 02:00
“4kV Bus U/V, 480 V Bus U/V, 125 VDC” received in the Main Control Room.

NARRATIVE

7/02/2024
Maintenance walked down 1A EDG looking for indications of what was causing the hanging alarm. Based on history and past experience, the battery charger/monitor was thought to be the most likely cause. No alarms on the battery charger were noted. No errors on the battery monitor noted and all measured points were within band. No abnormal indications noted on the 480v U/V relay.

7/03/2024
Maintenance calibrated the undervoltage relay with the dropped flag (127-1 UV relay). Calibration was performed satisfactorily with no adjustments needed. The issue appeared to be coming from the PT drawer, either a blown primary fuse or an issue with the PT itself.

7/05/2024
Discussions were held between the technicians and the FIN-SRO in which it was decided to implement a temporary configuration change for UV relay 127-1 to eliminate the hanging alarm.

7/08/2024
Print reviews indicated a 127 (undervoltage) relay pickup for control power and determined this is consistent with the UV alarm. The 127/151/B17 IFCV, Induction Disc Time-Overcurrent Relay with Voltage Restraint and Inverse Time Characteristics, type relay which was further evaluated and concluded that a loss of the ‘A’ phase would impact pickup amperage of the device, potentially tripping the diesel output breaker 152-1703 during an accident scenario. Maintenance and engineering provided this information to the Control Room.

TS Condition 3.8.1.B was entered for Unit 1 and TS Condition 3.8.1.E was entered for Unit 2 at 1315.

7/11/2024
Completed PT transformer replacement and verified proper voltage indication on all three phases. Performed a 1-hour loaded run for post maintenance operability test. The 1A EDG was restored to operable status at 0418 and the associated TS Conditions were exited.

D. CAUSE OF EVENT

The direct cause of the Potential Transformer failure is attributed to age-related degradation of the winding insulation causing turn-to-turn shorts on the primary and secondary winding. The turn-to-turn shorts caused increased currents on the primary side of the PT required to maintain secondary side voltage resulting in the upstream fuse on the primary side of PT-1 to overheat. The downstream effect was a loss of voltage to the 127-1 undervoltage relay, issuing the 1A EDG alarm, and the 127/151 voltage-restraint overcurrent relay, rendering the 1A EDG inoperable.

Contributing causes included gaps in determining the operability impact and a lack of independence during the initial operability determination.

NARRATIVE

E. SAFETY ANALYSIS

The subject event satisfies the criteria in NUREG-1022, Revision 3, for any operation or condition that was prohibited by the plant's technical specifications. The 4kV Bus U/V, 480 V Bus U/V, 125 VDC" received in the Main Control Room on 7/2/2024 at 2 am. The 1A EDG was declared inoperable on 7/8/2024, and the station entered TS Condition 3.8.1.B for Unit 1 and 3.8.1.E for Unit 2. After entering the TS Conditions the site took all requisite Required Actions. The 1A EDG was declared operable on July 11, 2024. However, prior to entering the TS Conditions and taking the associated requisite Required Actions on 7/8/2024, Unit 1 exceeded the completion time for TS Condition 3.8.1.B and the completion time for the subsequently Required Actions. Additionally, Unit 2 exceeded the completion time for TS Condition 3.8.1.E and the completion time for the subsequently Required Actions. Therefore, this event is reportable pursuant to 10CFR 50.73(a)(2)(i)(B). This event did not result in any actual nuclear safety consequences.

F. CORRECTIVE ACTIONS

The potential transformer was replaced on 7/10/2024, and following a successful surveillance run the 1A EDG was declared Operable on 7/11/2024. Training, observations, and procedure change actions have been taken to address the human performance factors that contributed to the event.

G. PREVIOUS OCCURRENCES

A review of previous Calvert Cliffs' events was performed for the last three years. The review did not identify any events stemming from the failure of a potential transformer which resulted in a condition prohibited by the plant's Technical Specifications.

H. COMPONENT FAILURE DATA

Component	IEEE 803 FUNCTION ID	IEEE 805 SYSTEM ID
Transformer, Potential.	XPT	EK